

①

Let X be a topological space.

Consider outcome relations of the following nature:

$$pRu, u \in C \subseteq \bar{U}, U \text{ open} \Rightarrow pRc.$$

There will be need to work on subsets of X as we will see.

~~② Keeping ① in mind, we will construct outcome~~

② It can be easily checked that GB need not be satisfied.

$$\text{Recalling } GB: x \circ (y \circ z) = (x \circ y) \circ z.$$

③ Keeping ① in mind, let us construct 'suitable outcome relations on 'suitable space'.

Let $(D, \vee, \wedge)$ be a distributive lattice.

Then the set of prime filters of D will form a topological space whose basic closed sets are $\hat{a}$.

$$\text{Recalling } \hat{a} = \{p \mid a \in p\} \subseteq \text{prime filters.}$$

④ Let $(D, \vee, \wedge)$ be a distributive lattice and Δ be a monotone map on D .

~~Before~~ We already have definition of Q_Δ given by Venema:

$$pQ_\Delta T \text{ iff } \Delta a \in p \text{ for all } a \in F_T.$$

$$\text{Recalling } F_T := \{a \in D \mid T \subseteq \hat{a}\}.$$

Let X : prime filter space.
 We will extract subsets of $\mathcal{Q}_D$:

Define $\mathcal{Q}'_D$ on the following ~~subset~~ subspace of X .

$$Y = \text{open sets } \cup \text{ closed sets } \cup \{c \in X \mid U \subseteq c \subseteq \bar{U}, U \text{ open}\}$$

(i) $p\mathcal{Q}'_D U, U \text{ open} \text{ iff } p\mathcal{Q}_D U.$

(ii) $p\mathcal{Q}'_D U, U \text{ closed} \text{ iff } p\mathcal{Q}_D U.$

(iii) $p\mathcal{Q}'_D c, c \text{ neither closed nor open}$

iff $p\mathcal{Q}_D U$ where $U \subseteq c \subseteq \bar{U}$ for some open set U .

Can be easily checked the 'defining feature' is ~~satisfied~~ satisfied.

① This relation $\mathcal{Q}'_D$ need not be monotonic on Y , since something(s) like these may arise:

• $U \subseteq \bar{U} \subseteq c, U \text{ open}, c \text{ neither open nor closed}$

$p\mathcal{Q}'_D U$ but $p\mathcal{Q}'_D c$, meaning $p\mathcal{Q}_D V$ for all

↓
 meaning

$p\mathcal{Q}_D U$

open sets V s.t. $V \subseteq c \subseteq \bar{V}$

More possibilities can lead to failure of total monotonicity. Above is just one example.

I am trying to come up with concrete counter-examples to validity this.

⑤ Now let us try to work through the proof of composition:

Let $M = (M, \vee, \wedge, -, \Diamond_g)_{g \in G}$ be a module over the game algebra $\mathcal{G} = (G, \vee, \wedge, -, \Diamond)$ and let g and h be arbitrary elements of G .

Then $Q'_{g \Diamond h} = Q'_g \circ Q'_h$.

Proof: Suppose $P(Q'_g \circ Q'_h)T$

$\therefore \exists U \in \gamma, P Q'_g U$ and $U Q'_h T$ for all $u \in U$.

Case 1: T is open/closed.

$U Q'_h T \Rightarrow U Q_h T$ (by definition of Q'_h).

Now consider an arbitrary element a such that $T \subseteq \hat{a}$, so, $\Diamond_h a \in u$ for all $u \in U$.

$\therefore U \subseteq (\Diamond_h a)^\wedge$.

$$\underline{U \text{ open/closed}}: p Q_g' U \Rightarrow p Q_g U$$

$$\therefore \Box_g \Box_h a \in p$$

$$\Box_g \Box_h a = \Box_{g \circ h} a \text{ since } M \text{ is a } \text{modular} \text{ module on the game algebra } \mathcal{G}.$$

$$\therefore \Box_{g \circ h} a \in p$$

$$\underline{U \text{ neither open, nor closed}}: \exists V \text{ open s.t. } V \subseteq U \subseteq \bar{V} \text{ and } p Q_g V.$$

$$\text{We have } U \subseteq (\Box_h a)^\wedge$$

$$\therefore V \subseteq (\Box_h a)^\wedge$$

$$\therefore \Box_g \Box_h a \in p.$$

$$\Box_g \Box_h a = \Box_{g \circ h} a.$$

$$\therefore \Box_{g \circ h} a \in p.$$

Since $a \in F_T$ was arbitrary, we have $p Q_{g \circ h}' T$.

Case 2: T is neither open, nor closed.

~~$\exists T_{\alpha} \text{ open s.t. } T_{\alpha} \subseteq T \subseteq \overline{T_{\alpha}} \text{ and } u \in \text{int } T_{\alpha}$~~

For all $u \in U$, $\exists T_{\alpha}$ open s.t. $T_{\alpha} \subseteq T \subseteq \overline{T_{\alpha}}$ and
 $u \in \text{int } T_{\alpha}$.

$\therefore u \in \text{int } (UT_{\alpha})$ [by monotonicity of int].
for all $u \in U$.

$\therefore u \in \text{int } (UT_{\alpha})$ for all $u \in U$ [since UT_{α} is open].

By Case 1, $p \in \text{int } (UT_{\alpha})$ [since UT_{α} is open].

Now, $T_{\alpha} \subseteq T \subseteq \overline{T_{\alpha}}$

$\Rightarrow UT_{\alpha} \subseteq T \subseteq U\overline{T_{\alpha}} \subseteq \overline{(UT_{\alpha})}$

$\therefore p \in \text{int } T$.

For the other direction, suppose $p \mathcal{Q}'_g \mathcal{Q}'_h T$.

Case 1: T is closed.

Then $p \mathcal{Q}_{g \circ h} T$.

We can copy Venema's proof to get a closed set U such that $p \mathcal{Q}_g U$ and $u \mathcal{Q}_h T$ for all $u \in U$.

The ' U ' Venema used was this:

$$U = \bigcap \{ (\mathcal{Q}_h b)^\wedge \mid b \in F_T \}.$$

~~Then~~ $\therefore p \mathcal{Q}'_g U$ and $u \mathcal{Q}'_h T$ for all $u \in U$.

$$\therefore p (\mathcal{Q}'_g \circ \mathcal{Q}'_h) T$$

Case 2: T is open.

~~Then $p \mathcal{Q}_{g \circ h} U$ and $p \mathcal{Q}_{g \circ h} \overline{T}$~~

Then $p \mathcal{Q}_{g \circ h} T$ and $p \mathcal{Q}_{g \circ h} \overline{T}$ by monotonicity of $\mathcal{Q}_{g \circ h}$.

By Case 1, there is a closed set U such that

$$p \mathcal{Q}_g U \text{ and } u \mathcal{Q}_h \overline{T} \text{ for all } u \in U.$$

$$u \mathcal{Q}_h \overline{T} \Rightarrow u \mathcal{Q}_h T$$

$$\therefore p \mathcal{Q}'_g U \text{ and } u \mathcal{Q}'_h T \\ \therefore p (\mathcal{Q}'_g \circ \mathcal{Q}'_h) T$$

Case 3: T is neither open, nor closed.

$\therefore \exists V$ open, $V \subseteq T \subseteq \overline{V}$ and $p Q_g \triangleleft_h V$.

$\therefore p Q_g \triangleleft_h \overline{V}$ by Monotonicity of $Q_g \triangleleft_h$.

By Case 1, $\exists$ closed set U st $p Q_g U$ and $u Q_h \overline{V}$.

~~$\therefore p Q_g' U$ and $u Q_h' \overline{V}$.~~

~~$u Q_h' V$~~ $u Q_h \overline{V} \Rightarrow u Q_h V$.

$\therefore p Q_g' U$ and $u Q_h' V$.

$\therefore p(Q_g' \circ Q_h') V$.

$\therefore p(Q_g' \circ Q_h') T$.